Eigenvalues

Introduction:

- It's just a number.
- This number may be of any type: +ve, -ve, zero, integer, fraction, irrational, even imaginary, complex.
- It's associated with square matrices.
- An *nxn* square matrix must have *n* eigenvalues, neither more, nor less.
- EVs of a matrix may be (all) distinct, (all) same, (some) repeated.
- A synonym for *eigenvalue* is *characteristic value*.

An all-covering question:

$$M = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$$

Answer the following questions listed on the RHS.

1. Find the ***characteristic matrix*** of M
2. Find the ***characteristic polynomial*** of M
3. Find the ***characteristic equation*** of M
4. Find the ***characteristic values*** of M
5. Find a spectrum of M
6. Classify M
7. Find |M|
8. Find a spectrum of M^T
9. Find a spectrum of M^3
10. Find a spectrum of M^{-1}
11. Find a spectrum of M^{-3}
12. Find the trace of M^{-3}
13. Find $|M^{-3}|$.

This Q covers all of the information about eigenvalues.

Theory: If λ be an *eigenvalue* (or, characteristic value) of M, then

1. **Answer:** $M - \lambda I_n$ is called the characteristic matrix of M ;

Therefore Characteristic matrix of M is,

$$M-\lambda I_3=\begin{bmatrix}4 & 6 & 6\\ 1 & 3 & 2\\ -1 & -4 & -3\end{bmatrix}-\lambda\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}=\begin{bmatrix}4-\lambda & 6 & 6\\ 1 & 3-\lambda & 2\\ -1 & -4 & -3-\lambda\end{bmatrix}.$$

2. **Answer:** $|M-\lambda I_n|$ is called the characteristic polynomial of M ;

   Therefore Characteristic polynomial of M is,

$$|M-\lambda I_3|=\begin{vmatrix}4-\lambda & 6 & 6\\ 1 & 3-\lambda & 2\\ -1 & -4 & -3-\lambda\end{vmatrix}=-\lambda^3+4\lambda^2+\lambda-4\,.$$

3. **Answer:** The characteristic equation of M is obtained by **putting zero equal** to the *characteristic polynomial* ; $\therefore |M-\lambda I_n|=0$

   Therefore Characteristic equation of M is,

$$|M-\lambda I_2|=-\lambda^3+4\lambda^2+\lambda-4=0$$

4. **Answer:** Roots of the characteristic equation are called the eigenvalues of M ;

   Therefore the eigenvalues are, (by solving the characteristic equation)

$$\therefore -\lambda^3+4\lambda^2+\lambda-4=0 \qquad \lambda=4,-1,1$$

5. **Answer:** A set of eigenvalues is called a spectrum of M ;

   Therefore the spectrum is,

$$\{-1,1,4\}$$

6. **Answer:** M is nonsingular ***iff*** all eigenvalues are nonzeroes but M is singular ***iff*** any one of eigenvalues are zeroes.

   Therefore

$$\lambda\neq 0$$
$$\Rightarrow \text{M is nonsingular.}$$

7. **Answer:** $|M|=$ (Product of eigenvalues of M) & tr(M)=Σ(eigenvalues of M).
   Therefore

$$|M|=(-1)(1)(4)=-4$$
$$tr(M)=-1+1+4=4$$

8. **Answer:** M & M^T posses same spectrums.

Therefore the spectrum of M^T is,

$$\{-1, 1, 4\}$$

Theory (contd):

❾ If M has an ev λ, then M^k has an ev λ^k, when $k = 2, 3, \ldots, r$;

❿ If M has ev λ & M is nonsingular, then $1/\lambda$ is an ev of M^{-1}.

⓫ The result follows when ❾ & ❿ are combined.

Solution (in progress):

❾ Here $k = 3$ & $\lambda = 4, -1, 1$

$\Rightarrow$ M^3 has evs $\lambda^3 = 4^3, (-1)^3, 1^3$

$\Rightarrow$ M^3 has a spectrum { 64, -1, 1 }.

❿ M is nonsingular [see ❻ at sld-7].

$\therefore$ M has ev : 4 , -1, 1

$\therefore$ spectrum of M^{-1} is {¼, -1, 1 }.

⓫ A spectrum of M^{-3} is { 1/64, -1, 1 }.

⓬ $tr(M^{-3}) = (1/64) + 1 + (-1) = 1/64$.

⓭ $|M^{-3}| = (1/64)(1)(-1) = -1/64$.

SQ

1. How many evs may a 13x13 matrix have ?
2. How many evs do belong to a 17x18 matrix ?
3. A singular matrix as a spectrum { -4, 1, 7 } : **True or False ?**
4. A nonsingular matrix must have only nonzero evs : **True or False ?**
5. Since tr(M) = Σ(evs of M) , then evs of M are diagonal elements of M : **True or False ?**

BQ

- Answer all questions mentioned in the sld-3, with respect to the following matrices:

a. $\begin{pmatrix} 4 & 3 \\ 2 & 5 \end{pmatrix}, \begin{pmatrix} 1 & 4 \\ 9 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix}$

b. $\begin{pmatrix} 5 & 7 & 1 \\ 0 & -2 & 6 \\ 0 & 0 & 3 \end{pmatrix}, \begin{pmatrix} 6 & 3 & -3 \\ -2 & -1 & 2 \\ 16 & 8 & -7 \end{pmatrix}, \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 1 \\ 2 & 1 & 0 \end{pmatrix}$

c. $\begin{pmatrix} 5 & 0 & 4 \\ 0 & 2 & 1 \\ 4 & 1 & 3 \end{pmatrix}, \begin{pmatrix} 3 & 4 & 0 \\ -4 & 5 & 9 \\ 0 & 8 & 6 \end{pmatrix}$

- In some cases, your answer may be undefined.
- In some cases c-numbers may occur as answers.